

Verifiable Federated Representation Learning for Cross-domain Sequential Recommendation

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Abstract

Cross-domain sequential recommendation (CDSR) plays a critical role in decentralized Web applications by leveraging user behavior sequences across multiple platforms to alleviate data sparsity and capture dynamic preferences. However, existing federated CDSR frameworks face two fundamental challenges: (i) heterogeneous sequential interactions that encode domain-exclusive semantics and cannot be directly shared under privacy constraints, and (ii) strong trust assumptions that both servers and clients behave honestly, leaving federated training vulnerable to misreporting, malicious updates, and negative transfer. In this paper, we propose **VeriFRL**, a verifiable federated representation learning framework for cross-domain sequential recommendation. VeriFRL adopts a dual-module design that integrates representation learning with verifiable training: an attention-based variational encoder disentangles domain-shared and domain-exclusive representations to support transferable and privacy-preserving knowledge sharing, while a contribution evaluation module quantifies client-level and feature-level influences to enable verifiability, interpretability, and negative transfer detection. Extensive experiments on real-world multi-domain datasets demonstrate that VeriFRL achieves competitive or superior recommendation performance over state-of-the-art federated CDSR methods, while providing fine-grained insights into cross-domain knowledge transfer dynamics.

CCS Concepts

• **Information systems** → **Web applications**; • **Computing methodologies** → **Distributed artificial intelligence**.

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Keywords

Federated Representation Learning; Contribution Evaluation; Semantic Knowledge Transfer; Verifiable Training

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1 Introduction

Cross-domain recommendation (CDR) has attracted considerable research interest due to its broad application scenarios on the Web, such as alleviating data sparsity and improving recommendation quality in e-commerce platforms or online services by leveraging auxiliary knowledge from other domains [27, 38]. Despite these advances, most existing efforts primarily focus on transferring static cross-domain knowledge (e.g., user/item embeddings or interaction patterns), while paying limited attention to the sequential nature of user behaviors that capture users' dynamic preferences [5, 34]. To address this gap, cross-domain sequential recommendation (CDSR) has emerged, aiming to exploit users' sequential interaction histories across multiple domains to enhance recommendation performance by modeling semantic correlations underlying behavioral dynamics [7, 9, 11, 30]. Building on these advances, prior CDSR studies have explored various strategies to further improve performance, including mixture-of-experts architectures, loss correction to mitigate negative transfer in centralized models [15, 25], and semantic fusion across domains [12, 35].

However, in real-world Web environments, user behavior data is naturally distributed across multiple platforms or service providers, each capturing different aspects of user preferences. Due to worldwide privacy regulations (e.g., GDPR and CCPA) and organizational policies, the raw data cannot be directly shared across domains, posing significant challenges for training effective CDR models. Recent

studies have explored privacy-preserving CDR frameworks that employ federated learning combined with differential privacy mechanisms to provide privacy guarantees during both cross-domain knowledge transfer and global model training [3, 10, 20, 22]. Although existing federated CDSR methods have shown initial success, some key challenges remain. The first challenge arises from **the heterogeneity of sequential user–item interactions under privacy-preserving settings**. These sequences encode domain-exclusive semantics of user behaviors that are highly sensitive and cannot be directly shared across domains. For instance, a user’s purchase sequence in one category may reveal narrative or brand-level preferences, while their interactions in another category reflect entirely different semantic patterns. This heterogeneity complicates cross-domain knowledge transfer and raises the question of how to preserve essential semantics while respecting privacy constraints.

The second challenge arises from **the strong trust assumptions in existing federated CDSR frameworks**. While most approaches leverage cross-domain information to improve recommendation performance in the target domain, they typically assume that both the central server and all participating clients behave honestly [37]. In practice, untrustworthy participants may submit manipulated or outdated model updates, or even inject adversarial gradients that distort the semantics of knowledge transfer and degrade the global model. Moreover, transferring knowledge from one domain to another can sometimes lead to negative transfer, where imported information harms rather than benefits the target application [33]. Despite efforts to mitigate these issues, current approaches lack verifiable mechanisms to ensure both the integrity of client behaviors and the semantic reliability of cross-domain knowledge transfer in federated CDSR. Recent work on client-level contribution measurement provides a promising direction, as it enables validation of client updates and offers new perspectives on evaluating the quality of semantic knowledge transfer in federated environments [4, 6, 19].

In this paper, we propose **VeriFRL**, a verifiable federated representation learning framework for cross-domain sequential recommendation. VeriFRL is composed of two key modules. First, to support interpretable and transferable representation learning, the *VAE-based Disentanglement Module* employs an attention-based variational encoder to model user interaction sequences and disentangle the latent space into domain-shared and domain-exclusive representations. The shared representation captures transferable cross-domain knowledge and participates in federated aggregation, while the exclusive representation preserves domain-specific preferences and is locally enhanced through contrastive augmentation, thereby alleviating the heterogeneity of sequential interactions in federated CDSR. Second, the *Contribution Evaluation Module* introduces verifiability into federated training. Inspired by TracIn [17], it traces the influence of training updates to estimate contributions at two levels: client-level (measuring each domain’s impact on the global model) and feature-level (quantifying the relative importance of shared vs. exclusive representations). This dual-level evaluation quantifies client- and feature-level contributions, making federated CDSR verifiable and interpretable, and enables the detection of negative transfer at both the domain and feature levels in real-world federated settings. We focus on verifiable training that audits the semantic alignment of each client update. Rather than

assuming honest participation, VeriFRL provides evidence to detect misreporting and negative transfer without requiring changes to the underlying FedAvg optimization. We summarize our major contribution as follows.

- We propose **VeriFRL**, a federated representation learning framework for cross-domain sequential recommendation that disentangles *domain-shared* and *domain-exclusive* semantic factors via an attention-based VAE, enabling privacy-preserving semantic transfer across domains.
- We introduce a *dual-level contribution evaluation* module that audits both client-level and feature-level influence, providing verifiable evidence of *semantic* transfer and exposing negative transfer when domain semantics are misaligned.
- Experiments on real-world multi-domain benchmarks show that VeriFRL achieves competitive or superior accuracy over strong federated CDSR baselines, and offers fine-grained semantic insights into when transfer helps or hurts under varying domain relatedness.

2 Related Work

2.1 Cross-Domain Sequential Recommendation

Early studies on CDSR primarily focused on dual-domain scenarios. Ma et al. [14] introduced π -Net, a parallel information-sharing network that exchanges representations across domains via shared-account user interaction sequences, and later proposed MIFN, which augments behavioral transfer with knowledge transfer over a cross-domain user–item graph to better capture users’ dynamic preferences [13]. Subsequent works extended CDSR with contrastive and graph-based techniques. Cao et al. [2] proposed a contrastive infomax objective to jointly model inter- and intra-domain interactions, while Xu et al. [31] introduced MGCL, a multi-view graph contrastive framework that integrates intra-domain item transitions and inter-domain user preferences, and further extended it to MGCL+ for multi-domain settings. More recently, large language models have been leveraged to exploit the semantic view of cross-domain information and enhance CDSR performance [11, 16, 29].

Despite these advances, most existing CDSR methods are developed under centralized training assumptions. Such assumptions are often impractical in real-world Web environments, where user data are naturally distributed across platforms and restricted by privacy regulations.

2.2 Federated Cross-domain Recommendation

In distributed systems, federated learning has been increasingly applied to cross-domain recommendation to preserve privacy while enabling effective knowledge transfer [21, 24, 32]. Early efforts mainly focused on semantic representation learning. For instance, Lu et al. [12] proposed FFMSR, which employs multi-layer semantic encodings and modality fusion to support non-overlapping scenarios. Later, contrastive learning was introduced into the federated setting: Wang et al. [24] proposed Fed-CLR, which integrates variational inference with generative reconstruction and consistency constraints to mitigate non-IID challenges, while Zheng et al. [40] proposed FedCSR, which applies dual contrastive learning to address both inter- and intra-platform heterogeneity without requiring user alignment. For specific applications, Wang et al. [23]

designed FL&PP-POI, which combines federated learning with homomorphic encryption to alleviate data sparsity and cold-start issues in POI recommendation. More recently, Zhang et al. [36] proposed FedDCSR, which disentangles domain-shared and domain-exclusive sequence representations and incorporates intra-domain contrastive infomax to better model heterogeneous user preferences in multi-domain scenarios. Yang et al. [33] further proposed FedGCDR, a federated graph-based framework that employs differential privacy-based knowledge transfer to mitigate privacy leakage and negative transfer in broader-source settings.

While these federated CDSR methods effectively address privacy preservation and cross-domain heterogeneity, they generally assume honest participation from both servers and clients. Model updates are treated as trustworthy signals, and the training process lacks explicit mechanisms to verify the integrity or semantic reliability of client contributions. As a result, existing approaches remain vulnerable to misreporting, malicious updates, and negative transfer, particularly in heterogeneous federated environments.

3 Methodology

3.1 Problem Definition

In this paper, we focus on cross-domain sequential recommendation under a federated learning paradigm. We consider a federation of K clients coordinated by a central server, where each client corresponds to a distinct domain (i.e., K equals the number of domains, and $K \geq 3$). Client k maintains its own user-item interaction dataset $\mathcal{D}_k = \{\mathcal{U}_k, \mathcal{S}_k^u\}_{u \in \mathcal{U}_k}$, where $\mathcal{U}_k$ is the user set within the domain k , and $\mathcal{S}_k^u = (s_1, s_2, \dots, s_T)$ denotes the chronological interaction sequence of user u . For representation learning, each client disentangles its sequences into domain-shared representations $\mathbf{Z}_k^s$, which capture transferable cross-domain knowledge, and domain-exclusive representations $\mathbf{Z}_k^e$, which preserve domain-specific preferences. After local training, the central server aggregates the shared representations $\{\mathbf{Z}_k^s\}_{k=1}^K$ to update the global model, which is then disseminated back to domain-level clients for the next training round.

3.2 Framework of VeriFRL

3.2.1 Overview. The overall framework of VeriFRL consists of three key modules: (i) a VAE-based disentanglement module for representation learning, (ii) a contribution evaluation module for verifiable training, and (iii) a federated optimization process for cross-domain collaboration. An overview of the architecture is illustrated in Figure 1. Following the VAE frameworks [36, 39], we assume a prior over latent variables and learn an approximate posterior conditioned on user-item sequences. Therewith, we directly build on this foundation and focus on disentangling domain-shared and domain-exclusive representations, as well as contribution evaluation in the federated setting.

3.2.2 VAE-based Disentanglement Module. Users in cross-domain recommendation exhibit both transferable and domain-specific preferences. To capture this duality, we disentangle the latent representation into a *domain-shared* component $\mathbf{z}_k^s$ and a *domain-exclusive* component $\mathbf{z}_k^e$. The shared part encodes transferable patterns and participates in federated aggregation, allowing clients to benefit

from overlapping user interaction behaviors across domains; the exclusive part preserves domain-unique semantics and remains local. As exclusive interactions are often sparse or noisy, we further enhance $\mathbf{z}_k^e$ with a contrastive information maximization (CIM) scheme [36]. We later quantify the respective impacts of the two components via our Contribution Evaluation Module.

Sequence encoder and variational posterior. Given the embedded sequence $\mathbf{E}_k \in \mathbb{R}^{T \times d}$, we employ a causal multi-head self-attention encoder [8] with residual connections, a position-wise feed-forward network, and LayerNorm, to obtain a sequence representation $\mathbf{h} \in \mathbb{R}^d$. (Causality is enforced by a mask $\mathbf{M}$ so that position i only attends to $\leq i$.) From $\mathbf{h}$ we parameterize a diagonal-Gaussian posterior:

$$\boldsymbol{\mu} = \mathbf{h}\mathbf{W}_\mu + \mathbf{b}_\mu, \quad \log \sigma^2 = \mathbf{h}\mathbf{W}_\sigma + \mathbf{b}_\sigma, \quad (1)$$

where $\boldsymbol{\mu}, \sigma \in \mathbb{R}^{d_z}$ denote the mean and standard deviation of $q_\phi(\mathbf{z} | \mathcal{S}_k^u)$. The latent variables can be sampled via a reparameterization process:

$$\mathbf{z} = \boldsymbol{\mu} + \sigma \odot \boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}), \quad (2)$$

where $\boldsymbol{\epsilon}$ denotes Gaussian noise ensuring differentiability of the sampling process.

Domain-shared knowledge representation. We instantiate a shared posterior distribution $q_\phi^s(\mathbf{z}_k^s | \mathcal{S}_k^u)$, modeled as a diagonal Gaussian whose mean and variance parameters are predicted from the encoder output $\mathbf{h}$. The domain-shared latent representation is obtained via the reparameterization trick:

$$\mathbf{z}_k^s = \boldsymbol{\mu}_k^s + \sigma_k^s \odot \boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}). \quad (3)$$

To encourage the shared encoder to capture transferable semantics across domains, we introduce a similarity loss $\mathcal{L}_k^{\text{similarity}}$ that maximizes the mutual information between the domain-shared representation $\mathbf{z}_k^s$ and the corresponding global representation $\mathbf{z}_k^g$:

$$\mathcal{L}_k^{\text{similarity}} = -I(\mathbf{z}_k^s; \mathbf{z}_k^g). \quad (4)$$

This objective aligns the latent subspace of local encoders with the global semantic space, promoting consistent knowledge transfer under federated training. In practice, directly estimating $I(\cdot; \cdot)$ is intractable, and we employ a variational approximation inspired by Deep InfoMax [1] to optimize its lower bound using a contrastive learning strategy. It is worthy noting that the shared-encoder parameters ϕ^s and the shared-domain representation $\mathbf{z}_k^s$ is uploaded for federated aggregation.

Domain-exclusive Knowledge Representation. Similar to domain-shared knowledge representation, the domain-exclusive posterior distribution $q_\phi^e(\mathbf{z}_k^e | \mathcal{S}_k^u)$, while the domain-exclusive representation $\mathbf{z}_k^e$ can be obtained using Equation (3). To strengthen domain-specific semantics, we adapt a contrastive information maximization as follows.

Each user-item interaction sequence $\mathcal{S}_k^u$ is randomly shuffled to obtain the augmented $\tilde{\mathcal{S}}_k^u$. Then, the augmented sequences representation $\tilde{\mathbf{z}}_k^e$ can be obtained by the exclusive encoder, where the last hidden state is used as the user embedding.

We then construct a domain-exclusive contrastive infomax loss $\mathcal{L}_k^{\text{CIM}}$ to maximize the agreement of the positive pair $(\mathbf{z}_k^e, \tilde{\mathbf{z}}_k^e)$ against

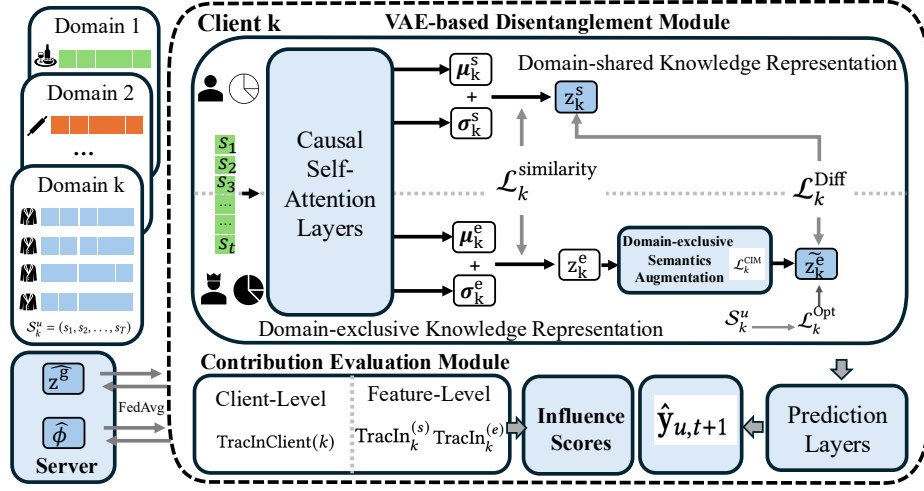


Figure 1: Overall architecture of the proposed VeriFRL framework.

negatives from other users in the mini-batch:

$$\mathcal{L}_k^{\text{CIM}} = -\log \frac{\exp(\cos(z_k^e, \tilde{z}_k^e)/\tau)}{\exp(\cos(z_k^e, \tilde{z}_k^e)/\tau) + \sum_{z^- \in \mathcal{Z}_k^-} \exp(\cos(z_k^e, z^-)/\tau)}, \quad (5)$$

where $\cos(\cdot, \cdot)$ is cosine similarity, τ is a temperature, and $\mathcal{Z}_k^-$ collects negatives in the batch. All exclusive-branch parameters (ψ, η) remain local and are never uploaded.

Therewith, we construct a domain-exclusive information optimization loss $\mathcal{L}_k^{\text{Opt}}$ to guide the sequential data representation reconstruction:

$$\mathcal{L}_k^{\text{Opt}} = -I(z_k^e; S_k), \quad (6)$$

where S_k denotes the user sequence within client k .

Since the mutual information term is intractable, we approximate it by maximizing its variational lower bound, which is implemented via a contrastive estimation objective that encourages z_k^e to retain domain-exclusive information relevant to S_k .

Similarly, we also construct a difference loss $\mathcal{L}_k^{\text{Diff}}$ to guide the disentanglement between the domain-shared and domain-exclusive representations. It aims to minimize the statistical dependency between z_k^s and z_k^e , encouraging the two latent spaces to capture complementary semantics. Formally, the loss can be formulated:

$$\mathcal{L}_k^{\text{Diff}} = I(z_k^s; z_k^e). \quad (7)$$

Training Objectives. For the local model update on the client k , the overall training objective $\mathcal{L}_k^{\text{Sum}}$ can be defined:

$$\mathcal{L}_k^{\text{Sum}} = \lambda_{\text{CIM}} \mathcal{L}_k^{\text{CIM}} + \lambda_{\text{Similarity}} \mathcal{L}_k^{\text{Similarity}} + \lambda_{\text{Opt}} \mathcal{L}_k^{\text{Opt}} + \lambda_{\text{Diff}} \mathcal{L}_k^{\text{Diff}}, \quad (8)$$

where λ are non-negative hyperparameters controlling the relative importance of each objective. Only the shared-branch parameters (ϕ^s) and the latent domain-shared representations z_k^s are uploaded for federated aggregation, while the exclusive-branch parameters (ϕ^e) and the exclusive representations z_k^e remain strictly local.

3.2.3 Contribution Evaluation Module. To make training verifiable, we estimate contributions at two levels: (i) client-level (domain)

influence on the global model, and (ii) feature-level importance of z_k^s vs. z_k^e for predictions. Considering the computational cost and efficiency, we adopt a TracIn-CP [17] influence score by tracing gradient inner products across checkpoints.

Client-level Contribution Evaluation. We adopt a TracIn-based influence estimation [17] to measure each client's contribution to the global model. The key idea is to check how well a client's local update aligns with the global validation objective during federated optimization. At training step t , client k produces a parameter update $\Delta\theta_k^{(t)}$ on the shared model parameters $\theta^{(t)}$. The client-level contribution is computed:

$$\text{TracInClient}(k) = \sum_{t \in \mathcal{T}} \langle \nabla_{\theta} \mathcal{L}_{\text{val}}(\theta^{(t)}), \Delta\theta_k^{(t)} \rangle, \quad (9)$$

where $\mathcal{L}_{\text{val}}(\cdot)$ is the validation loss and $\langle \cdot, \cdot \rangle$ denotes the inner product between the validation gradient and the local update. A higher score indicates that the client's update contributes positively to reducing the validation loss, thus benefiting the global model. Since the local learning rate η remains constant, we approximate the training gradient as $\nabla_{\theta} \mathcal{L}_{\text{train},k} \approx -\Delta\theta_k/\eta$, which allows efficient computation using parameter differences instead of full gradient evaluation.

It is important to note that in our framework, only the *domain-shared* representations and encoder parameters are uploaded for federated aggregation. As a result, the TracInClient score essentially measures the contribution of each client's **domain-shared knowledge**, indicating how much the shared semantic information learned within a local domain contributes to improving the global representation space. This formulation allows VeriFRL to interpret and verify the influence of each domain, distinguishing those that provide transferable semantic knowledge from those that may introduce negative transfer.

Feature-level Contribution Evaluation. For each prediction $\hat{y}_{u,t+1} = g_{\omega}(z_k^u)$, where $z_k^u = z_k^s + z_k^e$ represents the disentangled latent variables derived from the VAE module, $g_{\omega}(\cdot)$ denotes the prediction

head that operates on the normalized concatenation of the domain-shared and domain-exclusive representations.

To assess the individual contribution of each component, we estimate its marginal effect through gradient-based attribution:

$$\begin{aligned}\text{TracIn}_k^{(s)}(u, t) &= \left\langle \nabla_{z_k^s} \ell(\hat{y}_{u,t+1}, s_{t+1}), z_k^s \right\rangle, \\ \text{TracIn}_k^{(e)}(u, t) &= \left\langle \nabla_{z_k^e} \ell(\hat{y}_{u,t+1}, s_{t+1}), z_k^e \right\rangle,\end{aligned}\quad (10)$$

where $\ell(\cdot)$ denotes the prediction loss and the inner product measures the alignment between the gradient direction and the latent feature vector. These values indicate whether a feature component contributes to decreasing or increasing the prediction loss.

By aggregating over batches or checkpoints, we obtain cumulative estimates of component-wise influence:

$$\begin{aligned}\text{TracIn}_k^{(s)} &= \sum_{t \in \mathcal{T}} \left\langle \nabla_{z_k^s} \ell^{(t)}, \Delta z_k^{s(t)} \right\rangle, \\ \text{TracIn}_k^{(e)} &= \sum_{t \in \mathcal{T}} \left\langle \nabla_{z_k^e} \ell^{(t)}, \Delta z_k^{e(t)} \right\rangle.\end{aligned}\quad (11)$$

Operational interpretation and auditable client semantics. The prediction head consumes the fused latent representation $z_u^k = z_s^k + z_e^k$, where z_s^k captures transferable cross-domain semantics and z_e^k refines domain-specific preference. A positive and stable $\text{TracIn}_k^{(s)}$ indicates that shared semantics consistently improve the validation objective, whereas large or volatile $\text{TracIn}_k^{(e)}$ values reflect stronger personalization that can be more sensitive to sparsity and noise. By contrasting $\text{TracIn}_k^{(s)}$ and $\text{TracIn}_k^{(e)}$, VeriFRL provides a fine-grained view of whether transfer is beneficial or potentially negative. Moreover, client updates that are misaligned with the validation objective will exhibit weak or negative TracIn alignment, making semantic distortion and negative transfer observable even when the standard FedAvg aggregation rule is preserved.

Computational and communication overhead. The contribution evaluation is designed as an auditing layer rather than an additional optimization objective. In our implementation, TracIn-based evaluation is executed once every 5 communication rounds with 10 LiSSA iterations and damping 0.01 for stable Hessian–vector approximation. The computation mainly involves (i) one validation-gradient evaluation at selected checkpoints and (ii) inner products between the validation gradient and the uploaded shared-branch update $\Delta\theta_k$ (or latent-feature attribution for the feature-level score). Importantly, only shared-branch parameters are audited and transmitted, while the exclusive branch remains strictly local. On the communication side, the server receives the standard FedAvg update together with a small set of scalar contribution scores, and the aggregation rule itself remains unchanged, keeping the bandwidth overhead negligible compared to model updates.

3.2.4 Federated Optimization of VeriFRL. VeriFRL adopts a standard FedAvg-style optimization scheme, with an additional contribution evaluation step to enable verifiability. At the beginning of round r , the server maintains the global shared parameters $(\phi^s, \theta^s)^{(r)}$ and broadcasts them to all clients. Each client k then minimizes its local disentanglement objectives $\mathcal{L}_k^{\text{Diff}}$, $\mathcal{L}_k^{\text{Similarity}}$, and $\mathcal{L}_k^{\text{Opt}}$ (see Equation (8)) on $\mathcal{D}_k$, updating both the shared parameters

(ϕ^s, θ^s) and the exclusive parameters (ϕ^e, θ^e) , where the latter are always kept local.

After local training, clients compute client-level and feature-level contribution scores according to Equations (7), (6), (5). These scores are transmitted together with the shared updates $\Delta(\phi^s, \theta^s)$ to provide verifiable evidence of update quality and to detect potential misbehaving clients. In practice, these auditable signals can be used by the system operator for logging, alerting, or manual quarantine decisions, while this paper keeps the aggregation dynamics unchanged for a controlled comparison. Importantly, the contribution scores are currently not used to modify the aggregation rule, while the server aggregates updates following the standard FedAvg weighting by local data size n_k :

$$(\phi^s, \theta^s)^{(r+1)} = \sum_{k=1}^K \frac{n_k}{\sum_{j=1}^K n_j} ((\phi^s, \theta^s)^{(r)} + \Delta(\phi^s, \theta^s)_k). \quad (12)$$

This procedure ensures effective cross-domain knowledge transfer through federated aggregation, while the contribution evaluation mechanism provides verifiability and interpretability by revealing the relative influence of clients and representation components, without altering the optimization dynamics of FedAvg.

4 Experiments

4.1 Experimental Setup

4.1.1 Datasets. We evaluate VeriFRL on 10 popular domains from the real-world Amazon dataset, which has been widely adopted in cross-domain sequential recommendation research [14, 33, 36]. Following the preprocessing scheme in [36], we construct three cross-domain groups: (1) *Food–Kitchen–Clothing–Beauty* (FKCB), (2) *Movie–Book–Game* (MBG), and (3) *Sport–Garden–Home* (SGH).

Users and items with fewer than 10 interactions are filtered out. User interaction sequences are retained only if their lengths fall within [4, 16]. We split each user’s interactions chronologically, using the earliest 80% for training, the next 10% for validation, and the remaining 10% for testing. Detailed dataset statistics and domain-level distributions are reported in Appendix A.

4.1.2 Task Setting. We study the task of **cross-domain sequential recommendation** in a federated learning setting. A federation consists of K clients coordinated by a central server, where each client represents an independent domain (e.g., Food, Movie, or Book) with its own user–item interactions $\mathcal{D}_k = \{\mathcal{U}_k, \mathcal{S}_k^u\}_{u \in \mathcal{U}_k}$. Some users appear in multiple domains, forming transferable behavioral patterns, while others are domain-exclusive. The goal is to collaboratively learn a global model that enhances recommendation performance across all domains by sharing domain-shared representations z_k^s while keeping domain-exclusive representations z_k^e and raw data strictly local.

4.1.3 Evaluation Metrics. We adopt three standard metrics for evaluating top- N sequential recommendation performance: **MRR**, **HR@10**, and **NDCG@10**. Mean Reciprocal Rank (MRR) measures the average reciprocal rank of the ground-truth item in the predicted list, emphasizing the accuracy of the top-ranked results. Hit Rate (HR@10) calculates the proportion of cases where the true next item appears within the top 10 recommendations. Normalized Discounted Cumulative Gain (NDCG@10) further accounts for the

rank position of the correct item within the top 10, assigning higher weights to higher-ranked hits. It is noted that the higher values of all three metrics indicate better recommendation performance.

4.1.4 Baselines. We compare **VeriFRL** against several state-of-the-art sequential recommendation models. Since these methods are originally proposed in centralized settings, we adapt them to the federated scenario by applying FedAvg for model aggregation. The baselines are grouped into three categories:

- **Attention-based Models:** SASRec [8], a Transformer-based sequential recommender that captures long-range dependencies via self-attention. We adapt it to federated learning, denoted as **FedSASRec**.
- **VAE-based Models:** VSAN [39], which introduces variational self-attention networks to model stochasticity in user-item interactions, and ContrastVAE [26], which combines VAE with contrastive learning for robust representation learning. Their federated variants are denoted as **FedVSAN** and **FedContrastVAE**. Besides, we also compare our method with FedDCSR model [36], which extends the variational sequential recommendation framework to the federated setting by disentangling domain-shared and domain-specific representations.
- **Contrastive learning-based Models:** CL4SRec [28], which leverages cross-view contrastive objectives for sequence representation, and DuoRec [18], which employs dual contrastive learning for robustness. Their federated versions are denoted as **FedCL4SRec** and **FedDuoRec**.

4.1.5 Implementation Details. We implement all experiments using PyTorch 2.7.0 with CUDA support and NumPy 1.26.4 under Python 3.12.6. All experiments are conducted on an NVIDIA RTX3090 (24 GB). Each client is trained locally with a batch_size of 512 and a maximum sequence length of 16, where shorter interaction sequences are padded and longer ones are truncated. The model is optimized using the Adam optimizer with an initial learning rate of 0.001 and a weight decay of 0.0005. The global federated training runs for 40 communication rounds, and each client performs 3 local epochs per round. The learning rate decays by a factor of 0.1 every 10 epochs if no improvement is observed on the validation set. Early stopping is applied with a patience of 5 rounds. Following prior works [36, 39], we set the temperature of the contrastive loss to 1.0 and employ dot-product similarity.

For verifiable training, the TracIn-based contribution evaluation module is executed every 5 rounds, with 10 iterations in the LiSSA procedure and a damping coefficient of 0.01 to ensure stable Hessian-vector estimation. All random seeds are fixed to 42 for reproducibility. Code implementation of this paper is available at <https://doi.org/10.5281/zenodo.18315007>.

4.2 Recommendation Performance

We evaluate VeriFRL and baseline models on three cross-domain groups: Movies–Books–Games (MBG), Food–Kitchen–Clothing–Beauty (FKCB), and Sport–Garden–Home (SGH). The results are reported in Tables 1, 2, and 3 in terms of MRR, HR@10, and NDCG@10 (%). For each model, we additionally report the weighted average across domains, where weights are proportional to the number of

training interactions in each domain. This setting reflects realistic unbalanced-domain federated scenarios and provides a more reliable estimate of overall recommendation quality.

In the **Food–Kitchen–Clothing–Beauty** group (Table 2), where domains share strong semantic and functional correlations, VeriFRL attains the best overall performance (**MRR = 14.18%**, **HR@10 = 26.24%**, **NDCG@10 = 16.08%**), outperforming FedDCSR (**MRR = 12.70%**) and FedContrastVAE (**MRR = 9.48%**). The gains are particularly clear on the Kitchen and Beauty domains, where VeriFRL improves HR@10 by +3.37% and +2.05%, respectively. These results suggest that when domains exhibit coherent semantics (e.g., overlapping item taxonomies and consistent behavioral patterns), VeriFRL can effectively transfer shared representations while reducing interference from domain-exclusive noise. Moreover, the contribution evaluation module provides auditable evidence to distinguish transferable versus domain-specific signals, supporting more transparent and verifiable cross-domain learning.

In the **Movies–Books–Games** group (Table 1), VeriFRL achieves **8.14% MRR**, **16.48% HR@10**, and **9.16% NDCG@10**, remaining highly competitive with FedDuoRec (**9.33% MRR**). Although the improvement margins are narrower here, the results indicate that VeriFRL generalizes well in moderately related domains, where narrative or genre structures provide partial semantic overlap. In particular, the causal multi-head self-attention encoder enforces temporally consistent semantics in sequential representations, which helps stabilize knowledge transfer compared to standard attention-based variants.

In the **Sport–Garden–Home** group (Table 3), VeriFRL achieves **2.83% MRR**, **4.33% HR@10**, and **2.79% NDCG@10**, remaining competitive but not superior to all baselines. This outcome is consistent with the relatively weak semantic correlation among SGH domains, where user intents and item entities differ substantially (e.g., outdoor sports equipment versus home decoration), making shared semantic alignment intrinsically harder. Nevertheless, VeriFRL still offers auditable signals indicating when transfer is beneficial versus potentially harmful, which is valuable for transparent deployment in heterogeneous federated environments. Overall, the results suggest that the degree of semantic relatedness plays a key role in determining the effectiveness of federated cross-domain transfer.

Finally, VeriFRL maintains stable accuracy even when domain-exclusive samples are sparse. For example, in the Beauty and Kitchen domains (with relatively fewer exclusive users), VeriFRL still surpasses VAE-based and attention-based baselines, suggesting that shared knowledge transfer implicitly regularizes underrepresented domains without privacy leakage.

In summary, VeriFRL achieves the strongest gains in semantically coherent settings and maintains competitive accuracy under heterogeneous or weakly aligned domains, while providing a verifiable and interpretable perspective on cross-domain knowledge transfer.

4.3 Verifiable and Explainable Contribution Evaluation

To examine the *verifiability* and *explainability* of **VeriFRL**, we analyze the contribution evaluation results from two complementary

Table 1: Performance comparison of federated sequential recommendation models on the MBG domains.

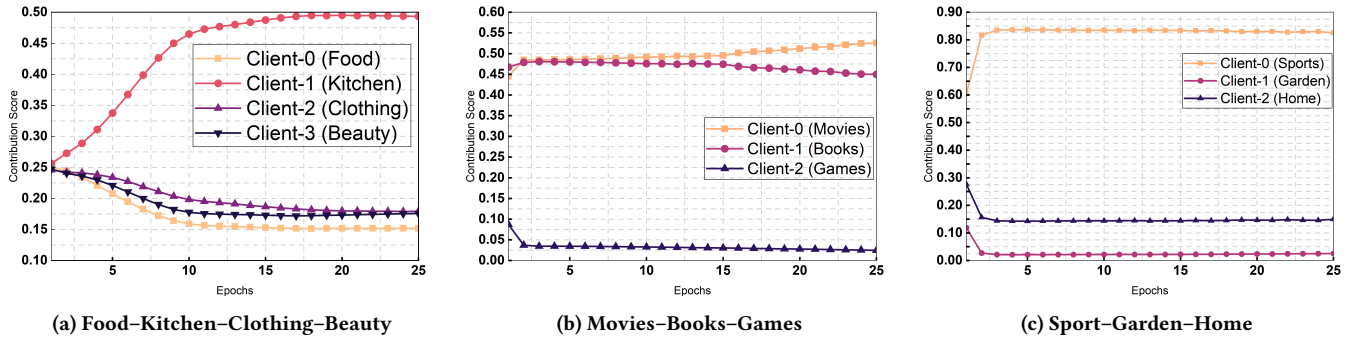
Model	Movie			Book			Game			Weighted Avg.		
	MRR	HR@10	NDCG@10	MRR	HR@10	NDCG@10	MRR	HR@10	NDCG@10	MRR	HR@10	NDCG@10
FedSASRec	9.19	15.30	9.90	5.66	8.42	5.78	5.13	6.65	5.16	7.23	11.43	7.61
FedVSAN	5.95	12.29	6.34	5.32	10.28	5.76	1.61	2.77	1.49	5.41	10.80	5.80
FedContrastVAE	9.08	16.33	9.99	5.62	10.26	6.00	5.36	6.44	5.29	7.17	12.80	7.76
FedCL4SRec	8.89	14.56	9.51	5.55	8.45	5.70	5.30	6.65	5.30	7.04	11.11	7.40
FedDuoRec	11.43	19.67	12.47	7.88	12.87	8.38	4.89	6.51	4.91	9.33	15.61	10.05
FedDCSR	6.00	12.75	6.49	5.42	10.51	5.88	7.21	11.36	7.63	5.78	11.57	6.25
VeriFRL (ours)	9.15	19.10	10.38	7.18	14.43	8.04	8.60	13.39	9.20	8.14	16.48	9.16

Table 2: Performance comparison of federated sequential recommendation models on the FKCB domains.

Model	Food			Kitchen			Clothing			Beauty			Weighted Avg.		
	MRR	HR@10	NDCG@10	MRR	HR@10	NDCG@10	MRR	HR@10	NDCG@10	MRR	HR@10	NDCG@10	MRR	HR@10	NDCG@10
FedSASRec	20.71	32.21	22.89	5.20	8.92	5.47	1.35	2.42	1.22	12.00	20.70	13.51	8.48	13.97	9.25
FedVSAN	17.96	33.48	20.74	5.17	10.03	5.52	1.91	4.16	1.92	8.28	18.95	9.83	7.42	14.74	8.36
FedContrastVAE	21.62	36.26	24.50	6.40	11.36	6.86	1.57	3.00	1.47	13.54	23.27	15.15	9.48	16.34	10.47
FedCL4SRec	20.92	32.51	23.14	5.00	9.18	5.38	1.30	2.08	1.10	12.13	20.12	13.48	8.45	13.96	9.22
FedDuoRec	20.13	31.61	22.28	5.23	8.96	5.50	1.53	2.19	1.32	12.38	19.42	13.47	8.49	13.60	9.16
FedDCSR	27.57	47.15	31.68	9.08	18.32	10.38	2.45	4.16	2.45	18.04	34.62	21.45	12.70	23.56	14.62
VeriFRL (ours)	28.15	45.12	31.64	10.99	21.69	12.85	1.96	3.07	1.85	18.18	32.67	21.03	14.18	26.24	16.08

Table 3: Performance comparison of federated sequential recommendation models on the SGH domains.

Model	Sport			Garden			Home			Weighted Avg.		
	MRR	HR@10	NDCG@10	MRR	HR@10	NDCG@10	MRR	HR@10	NDCG@10	MRR	HR@10	NDCG@10
FedSASRec	3.42	4.19	3.26	4.88	5.33	4.67	3.03	3.69	2.84	3.91	4.42	3.53
FedVSAN	1.21	2.24	1.05	1.22	1.90	1.02	1.37	2.46	1.23	1.39	2.13	1.15
FedContrastVAE	3.30	3.97	3.09	5.03	5.69	4.82	3.07	3.75	2.88	3.78	4.47	3.63
FedCL4SRec	3.43	4.20	3.28	4.74	5.59	4.61	2.95	3.71	2.77	3.69	4.38	3.47
FedDuoRec	3.77	4.52	3.61	4.76	5.30	4.55	3.29	3.80	3.05	3.79	4.46	3.58
FedDCSR	1.52	2.78	1.42	2.26	3.82	2.23	1.56	2.81	1.45	1.80	2.97	1.76
VeriFRL (ours)	1.92	3.48	1.82	6.51	8.66	6.59	1.78	3.18	1.67	2.83	4.33	2.79

**Figure 2: Client-level contribution evolution across three domain groups in the first 25 global rounds. Each curve shows the normalized contribution scores of clients over training epochs.**

perspectives: the *client level* and the *feature level*. These analyses provide interpretable evidence of how semantic knowledge is transferred across domains and how federated collaboration evolves during the model training process.

Client-level contribution. Each domain is treated as a client, and its influence on the global model is quantified using the TraIn-CP metric. Figure 3 shows the normalized contribution trajectories

across the first 25 global rounds. In the **FKCB** group, the Kitchen domain shows a clear upward trend, increasing from about 0.26 to 0.49, while the other three domains gradually stabilize around 0.17–0.19. This pattern indicates that Kitchen contains richer and more balanced interaction data, resulting in stronger semantic overlap with the other domains. In the **MBG** group, the Movies and Books domains contribute steadily between 0.45 and 0.52, while

Games remains below 0.03 due to limited shared semantics and sparse activity. In contrast, the **SGH** group exhibits the most uneven pattern, with Sports maintaining over 0.8 of total influence throughout training, while Garden and Home stay below 0.15. These observations show that domains with denser user–item interactions and higher semantic coherence naturally dominate the global optimization process. VeriFRL thus provides direct, verifiable evidence of inter-domain collaboration and how data richness and semantic alignment shape federated contribution dynamics.

Table 4: Feature-level contribution evaluation on the Food–Kitchen–Clothing–Beauty group.

Epoch	Client 0 (Food)		Client 1 (Kitchen)		Client 2 (Clothing)		Client 3 (Beauty)	
	Shared	Exclusive	Shared	Exclusive	Shared	Exclusive	Shared	Exclusive
5	-0.0138	-0.0432	-0.1738	-0.1549	-0.0466	0.6150	-0.0478	-0.1077
10	-0.0139	0.0098	0.0350	0.1518	-0.0366	0.6858	-0.0226	0.2428
15	0.0352	0.2364	0.0695	0.4843	0.0638	0.2772	0.0650	0.4014
20	0.0488	0.5866	0.0889	0.9999	0.1093	0.4652	0.0681	1.0074
25	0.0424	0.8400	0.0915	1.5406	0.1079	0.7678	0.0784	1.4557

Feature-level contribution. We investigate the internal behavior of VeriFRL by examining feature-level TraIn scores, which capture the influence of domain-shared and domain-exclusive representations. In the Food–Kitchen–Clothing–Beauty group, the early training rounds (5–10) exhibit unstable and negative scores, reflecting the stochastic stage before semantic alignment. From round 15 onward, shared representations become consistently positive and stable. For example, $\text{client}_1^{\text{shared}}$ increases from 0.07 to 0.09.

From a predictive perspective, this stable positive alignment indicates that domain-shared representations contribute directly to correct next-item prediction by capturing transferable cross-domain semantic patterns. In contrast, exclusive representations exhibit larger but more volatile TraIn values, such as $\text{client}_1^{\text{exclusive}}$ reaching 1.54 in round 25. This behavior suggests that exclusive features primarily capture strong personalization effects and local user preferences, while being more sensitive to sparse data and domain-specific noise. Overall, shared representations act as a stable semantic foundation that supports cross-domain generalization, whereas exclusive representations mainly refine local preferences.

Interpretation and implications. These results demonstrate that VeriFRL can provide interpretable and evidence-based insights into how federated models evolve across domains. Instead of masking or heuristically removing potential negative transfer, VeriFRL explicitly measures and visualizes the contribution of each client and feature type. This makes the optimization process more transparent and allows practitioners to understand which domains drive or hinder global improvement. In practice, such signals can support debugging and monitoring in federated deployments, for example by flagging persistently low or negative contributors for closer inspection.

From a semantic perspective, this interpretability is crucial for building trustworthy cross-domain systems on the Web. By quantifying the contribution of shared and exclusive knowledge components, VeriFRL links model behavior to the underlying semantic relationships among domains. As a result, it supports more accountable and verifiable collaboration, where participants can trace

how their data influence the collective representation space without revealing private information. This is particularly relevant in privacy-constrained settings where direct access to cross-domain interactions is unavailable and semantic alignment must be inferred from model dynamics. Overall, VeriFRL moves federated recommendation toward a more semantically interpretable paradigm, aligning federated optimization with transparent knowledge sharing in decentralized Web environments.

To further verify the interpretability and sensitivity of the proposed contribution evaluation module, we conduct a leave-one-domain-out ablation within the Food–Kitchen–Clothing–Beauty group. Details are reported in Appendix B.

5 Limitations and Future Work

VeriFRL is most effective when domains share overlap. When relatedness is weak, aligning a shared latent space becomes difficult and the benefit of knowledge transfer may be limited, with a higher risk of negative transfer under strong heterogeneity. In addition, the current TraIn-based contribution evaluation introduces auditing cost due to checkpointing and gradient-based influence approximation, even though it is computed post-hoc and does not modify the optimization objective or add training-time computation.

Future work can further explore two promising directions. First, incorporating the two-level contribution signals (client-level and feature-level) into adaptive learning pipelines may improve robustness and efficiency, e.g., via dynamic reweighting, resource-aware client prioritization, and selective aggregation of informative feature subspaces under privacy constraints. Second, from a Web perspective, auditable contribution signals may serve as a practical audit trail for accountable knowledge transfer in decentralized services, supporting monitoring, governance, and incentive-aware participation without exposing raw user data.

6 Conclusion

This work introduced **VeriFRL**, a verifiable federated representation learning framework for cross-domain sequential recommendation. VeriFRL disentangles user sequences into domain-shared and domain-exclusive factors via a VAE-based encoder, enabling privacy-preserving knowledge transfer while retaining domain-specific behaviors. It further provides a dual-level contribution evaluation module (client- and feature-level) that offers verifiable and interpretable evidence of cross-domain collaboration. Experiments on Web-scale multi-domain benchmarks show that VeriFRL improves recommendation quality by leveraging cross-domain relatedness, remains competitive under weakly related settings, and reveals when transfer is beneficial or potentially harmful. Overall, VeriFRL advances accountable knowledge sharing in decentralized Web systems without exposing sensitive data.

Acknowledgments

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References

- [1] Mohamed Ishmael Belghazi, Aristide Baratin, Sai Rajeshwar, Sherjil Ozair, Yoshua Bengio, Aaron Courville, and Devon Hjelm. 2018. Mutual Information Neural Estimation. In *Proceedings of the 35th International Conference on Machine Learning*, Vol. 80. 531–540.
- [2] Jiangxia Cao, Xin Cong, Jiawei Sheng, Tingwen Liu, and Bin Wang. 2022. Contrastive cross-domain sequential recommendation. In *Proceedings of the 31st ACM International Conference on Information & Knowledge Management*. 138–147.
- [3] Gaode Chen, Xinghua Zhang, Yijun Su, Yantong Lai, Ji Xiang, Junbo Zhang, and Yu Zheng. 2023. Win-Win: A Privacy-Preserving Federated Framework for Dual-Target Cross-Domain Recommendation. *Proceedings of the AAAI Conference on Artificial Intelligence* 37, 4 (2023), 4149–4156.
- [4] Yue Cui, Chung-ju Huang, Yuzhu Zhang, Leye Wang, Lixin Fan, Xiaofang Zhou, and Qiang Yang. 2024. A survey on contribution evaluation in vertical federated learning. *arXiv preprint arXiv:2405.02364* (2024).
- [5] Zhe Fu, Xi Niu, and Mary Lou Maher. 2023. Deep Learning Models for Serendipity Recommendations: A Survey and New Perspectives. *ACM Comput. Surv.* 56, 1, Article 19 (2023).
- [6] Jingjing Guo, Jiaxing Li, Zhiquan Liu, Yupeng Xiong, Yong Ma, Athanasios V. Vasilakos, Xinghua Li, and Jianfeng Ma. 2025. LCEFL: A Lightweight Contribution Evaluation Approach for Federated Learning. *IEEE Transactions on Mobile Computing* 24, 7 (2025), 6643–6657.
- [7] Clark Mingxuan Ju, Leonardo Neves, Bhuvish Kumar, Liam Collins, Tong Zhao, Yuwei Qiu, Qing Dou, Sohail Nizam, Sen Yang, and Neil Shah. 2025. Revisiting Self-attention for Cross-domain Sequential Recommendation. In *Proceedings of the 31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining V. 2*. 1094–1105.
- [8] Wang-Cheng Kang and Julian McAuley. 2018. Self-Attentive Sequential Recommendation. In *2018 IEEE International Conference on Data Mining (ICDM)*. 197–206.
- [9] Guanyu Lin, Chen Gao, Yu Zheng, Jianxin Chang, Yanan Niu, Yang Song, Kun Gai, Zhiheng Li, Depeng Jin, Yong Li, and Meng Wang. 2024. Mixed Attention Network for Cross-domain Sequential Recommendation. In *Proceedings of the 17th ACM International Conference on Web Search and Data Mining*. 405–413.
- [10] Zhenghong Lin, Wei Huang, Hengyu Zhang, Jiayu Xu, Weiming Liu, Xinting Liao, Fan Wang, Shiping Wang, and Yanchao Tan. 2024. Enhancing dual-target cross-domain recommendation with federated privacy-preserving learning. In *Proceedings of the 33rd International Joint Conference on Artificial Intelligence*. 2153–2161.
- [11] Qidong Liu, Xiangyu Zhao, Yejing Wang, Zijian Zhang, Howard Zhong, Chong Chen, Xiang Li, Wei Huang, and Feng Tian. 2025. Bridge the Domains: Large Language Models Enhanced Cross-domain Sequential Recommendation. In *Proceedings of the 48th International ACM SIGIR Conference on Research and Development in Information Retrieval*. 1582–1592.
- [12] Ziang Lu, Lei Guo, Xu Yu, Zhiyong Cheng, Xiaohui Han, and Lei Zhu. 2025. Federated Semantic Learning for Privacy-preserving Cross-domain Recommendation. *ACM Trans. Inf. Syst.* 43, 4, Article 103 (2025).
- [13] Muiyang Ma, Pengjie Ren, Zhumin Chen, Zhaochun Ren, Lifan Zhao, Peiyu Liu, Jun Ma, and Maarten de Rijke. 2022. Mixed Information Flow for Cross-Domain Sequential Recommendations. *ACM Transactions on Knowledge Discovery from Data* 16, 4 (2022).
- [14] Muiyang Ma, Pengjie Ren, Yujie Lin, Zhumin Chen, Jun Ma, and Maarten de Rijke. 2019. π -Net: A Parallel Information-sharing Network for Shared-account Cross-domain Sequential Recommendations. In *Proceedings of the 42nd International ACM SIGIR Conference on Research and Development in Information Retrieval* (Paris, France). 685–694.
- [15] Chung Park, Taesan Kim, Hyungjun Yoon, Junui Hong, Yelim Yu, Mincheol Cho, Minsung Choi, and Jaegul Choo. 2024. Pacer and Runner: Cooperative Learning Framework between Single- and Cross-Domain Sequential Recommendation. In *Proceedings of the 47th International ACM SIGIR Conference on Research and Development in Information Retrieval*. 2071–2080.
- [16] Alessandro Petruzzelli, Cataldo Musto, Lucrezia Laraspata, Ivan Rinaldi, Marco de Gemmis, Pasquale Lops, and Giovanni Semeraro. 2024. Instructing and Prompting Large Language Models for Explainable Cross-domain Recommendations. In *Proceedings of the 18th ACM Conference on Recommender Systems*. 298–308.
- [17] Garima Pruthi, Frederick Liu, Satyen Kale, and Mukund Sundararajan. 2020. Estimating training data influence by tracing gradient descent. *Advances in Neural Information Processing Systems* 33 (2020), 19920–19930.
- [18] Ruihong Qiu, Zi Huang, Hongzhi Yin, and Zijian Wang. 2022. Contrastive Learning for Representation Degeneration Problem in Sequential Recommendation. In *Proceedings of the Fifteenth ACM International Conference on Web Search and Data Mining*. 813–823.
- [19] Tao Tang, Mingliang Hou, Shuo Yu, Zhen Cai, Zhiwen Han, Giles Oatley, and Vidya Saikrishna. 2024. FedGST: An Efficient Federated Graph Neural Network for Spatio-temporal PoI Recommendation. *ACM Trans. Sen. Netw.* (2024).
- [20] Changxin Tian, Yuexiang Xie, Xu Chen, Yaliang Li, and Xin Zhao. 2024. Privacy-preserving Cross-domain Recommendation with Federated Graph Learning. *ACM Trans. Inf. Syst.* 42, 5, Article 135 (2024).
- [21] Cheng Wang, Wenchao Xu, Haozhao Wang, Wei Liu, and Ruixuan Li. 2025. Privacy-Friendly Cross-Domain Recommendation via Distilling User-irrelevant Information. In *Proceedings of the ACM on Web Conference 2025*. 450–461.
- [22] Li Wang, Shoujin Wang, Quanguai Zhang, Qiang Wu, and Min Xu. 2025. Federated User Preference Modeling for Privacy-Preserving Cross-Domain Recommendation. *IEEE Transactions on Multimedia* 27 (2025), 5324–5336. doi:10.1109/TMM.2025.3543106
- [23] Li-e Wang, Yihui Wang, Yan Bai, Peng Liu, and Xianxian Li. 2021. POI Recommendation with Federated Learning and Privacy Preserving in Cross Domain Recommendation. In *IEEE INFOCOM 2021 - IEEE Conference on Computer Communications Workshops (INFOCOM WKSHOPS)*. 1–6.
- [24] Qingren Wang, Yuchuan Zhao, Yi Zhang, Yiwen Zhang, Shuiguang Deng, and Yun Yang. 2025. Federated Contrastive Learning for Cross-Domain Recommendation. *IEEE Transactions on Services Computing* 18, 2 (2025), 812–827.
- [25] Shuliang Wang, Jiabao Zhu, Yi Wang, Chen Ma, Xin Zhao, Yansen Zhang, Ziqiang Yuan, and Sijie Ruan. 2025. Hierarchical gating network for cross-domain sequential recommendation. *ACM Transactions on Information Systems* 43, 4 (2025), 1–32.
- [26] Yu Wang, Hengrui Zhang, Zhiwei Liu, Liangwei Yang, and Philip S. Yu. 2022. ContrastVAE: Contrastive Variational AutoEncoder for Sequential Recommendation. In *Proceedings of the 31st ACM International Conference on Information & Knowledge Management*. 2056–2066.
- [27] Yongxuan Wu, Yang Liu, Xixun Lin, Hong Zhou, Yanan Cao, Lixin Zou, Yanmin Shang, and Yanbing Liu. 2025. FairCDR: transferring fairness and user preferences for cross-domain recommendation. In *Proceedings of the 31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining V. 2*. 3261–3272.
- [28] Xu Xie, Fei Sun, Zhaoyang Liu, Shiwen Wu, Jinyang Gao, Jiandong Zhang, Bolin Ding, and Bin Cui. 2022. Contrastive Learning for Sequential Recommendation. In *2022 IEEE 38th International Conference on Data Engineering (ICDE)*. 1259–1273.
- [29] Haoran Xin, Ying Sun, Chao Wang, and Hui Xiong. 2025. LLMCDSR: Enhancing Cross-Domain Sequential Recommendation with Large Language Models. *ACM Trans. Inf. Syst.* 43, 5, Article 120 (2025), 33 pages.
- [30] Wujiang Xu, Qitian Wu, Runzhong Wang, Mingming Ha, Qiongxi Ma, Linxun Chen, Bing Han, and Junchi Yan. 2024. Rethinking Cross-Domain Sequential Recommendation under Open-World Assumptions. In *Proceedings of the ACM Web Conference 2024*. 3173–3184.
- [31] Zitao Xu, Shu Chen, Weike Pan, and Zhong Ming. 2025. A Multi-view Graph Contrastive Learning Framework for Cross-Domain Sequential Recommendation. *ACM Trans. Recomm. Syst.* 3, 4 (2025).
- [32] Dengcheng Yan, Yuchuan Zhao, Zhongxiu Yang, Ying Jin, and Yiwen Zhang. 2022. FedCDR: Privacy-preserving federated cross-domain recommendation. *Digital Communications and Networks* 8, 4 (2022), 552–560.
- [33] Ziqi Yang, Zhaopeng Peng, Zihui Wang, Jianzhong Qi, Chaochao Chen, Weike Pan, Chenglu Wen, Cheng Wang, and Xiaoliang Fan. 2024. Federated Graph Learning for Cross-Domain Recommendation. In *Advances in Neural Information Processing Systems*, Vol. 37. 64865–64888.
- [34] Tianzi Zang, Yanmin Zhu, Haobing Liu, Ruohan Zhang, and Jiadi Yu. 2022. A Survey on Cross-domain Recommendation: Taxonomies, Methods, and Future Directions. *ACM Trans. Inf. Syst.* 41, 2, Article 42 (2022).
- [35] Yongfu Zha, Xinxin Dong, Haokai Ma, Yonghui Yang, and Xiaodong Wang. 2025. Align-for-Fusion: Harmonizing Triple Preferences via Dual-oriented Diffusion for Cross-domain Sequential Recommendation. *arXiv preprint arXiv:2508.05074* (2025).
- [36] Hongyu Zhang, Dongyi Zheng, Xu Yang, Jiyuan Feng, and Qing Liao. 2024. FedDCSR: Federated cross-domain sequential recommendation via disentangled representation learning. In *Proceedings of the 2024 SIAM International Conference on Data Mining (SDM)*. SIAM, 535–543.
- [37] Yanci Zhang and Han Yu. 2022. Towards Verifiable Federated Learning. In *Proceedings of the Thirty-First International Conference on Artificial Intelligence, IJCAI-22*. 5686–5693. Survey Track.
- [38] Guoshuai Zhao, Xiaolong Zhang, Hao Tang, Jialie Shen, and Xueming Qian. 2024. Domain-oriented knowledge transfer for cross-domain recommendation. *IEEE Transactions on Multimedia* 26 (2024), 9539–9550.
- [39] Jing Zhao, Pengpeng Zhao, Lei Zhao, Yanchi Liu, Victor S. Sheng, and Xiaofang Zhou. 2021. Variational Self-attention Network for Sequential Recommendation. In *2021 IEEE 37th International Conference on Data Engineering (ICDE)*. 1559–1570.
- [40] Dongyi Zheng, Hongyu Zhang, Jianyang Zhai, Lin Zhong, Lingzhi Wang, Jiyuan Feng, Xiangke Liao, Yonghong Tian, Nong Xiao, and Qing Liao. 2025. FedCSR: A Federated Framework for Multi-Platform Cross-Domain Sequential Recommendation with Dual Contrastive Learning. In *Proceedings of the 31st International Conference on Computational Linguistics*. 8699–8713.

Table 5: Statistics of the Amazon datasets.

Statistic	Food	Kitchen	Clothing	Beauty	Movie	Book	Game	Sport	Garden	Home
Total Users	4658	13382	9240	5902	34792	19419	5588	28139	6852	20784
Total Items	13564	32918	34909	17780	44464	72246	10336	88992	21604	62499
Training Set	4977	11100	5720	4668	57405	63157	6631	51477	10479	37361
Valid Set	1307	2172	818	836	10944	11168	1374	13720	3074	10058
Testing Set	1332	2254	866	855	11654	12149	1444	14214	3113	10421
Average Lenth	8.79	8.60	9.30	10.29	7.97	7.20	6.49	10.65	9.48	10.41

A Dataset Statistics and Domain Overlap

For a better understanding of the cross-domain overlap, we also analyze shared and exclusive users among these domain groups. Specifically, in the *Food–Kitchen–Clothing–Beauty* group, 1,348 users are shared across domains, while each domain has 97, 4,176, 630, and 146 exclusive users, respectively. In the *Movie–Book–Game* group, 1,669 users are shared, and the domains have 13,704, 3,641, and 192 exclusive users. For the *Sport–Garden–Home* group, 6,138 users are shared across domains, with 7,627, 11, and 452 exclusive users, respectively. These observations highlight varying degrees of user overlap and heterogeneity among domains, providing a realistic benchmark for evaluating federated cross-domain recommendation models.

B Ablation Study

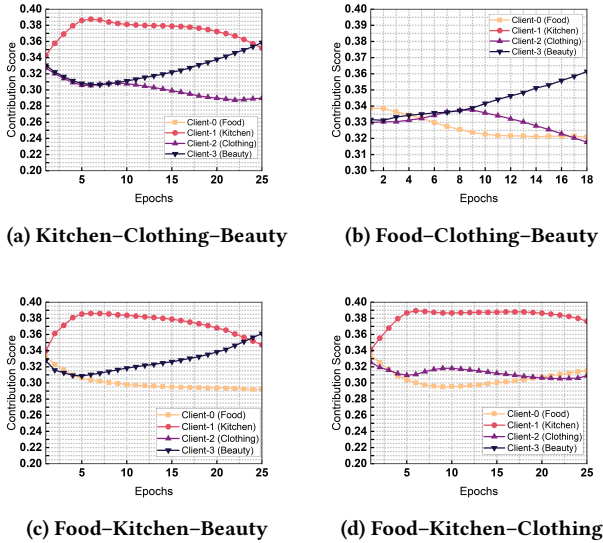


Figure 3: Client-level contribution evolution across domain groups. Each curve shows the normalized TracIn-based contribution scores over 25 training epochs. The last subfigure visualizes ablation results when excluding specific domains.

To further verify the interpretability and sensitivity of the proposed contribution evaluation module, we conduct a leave-one-domain-out ablation within the *Food–Kitchen–Clothing–Beauty* group.

We analyze the four ablation settings in Figure 3a–3d by relating the TracIn trajectories to the underlying data statistics in Table 5. All curves are normalized within each setting, therefore values sum to one across the active clients at a given epoch.

Kitchen–Clothing–Beauty (without Food). At epoch 1 the contributions are already close, with Kitchen at 0.3426, Clothing at 0.3270, and Beauty at 0.3304. As training proceeds, Kitchen first rises to about 0.3876 around epoch 6, then gradually declines to 0.3517 by epoch 25. Beauty steadily increases to 0.3586, while Clothing decreases to 0.2897. Removing Food mainly redistributes influence toward Beauty and, to a lesser extent, Kitchen. This behavior is consistent with Beauty’s relatively long average sequence length (10.29) that supports stronger sequential regularities, and with Kitchen’s larger training size, which sustains early dominance before balancing with Beauty at convergence.

Food–Clothing–Beauty (without Kitchen). Kitchen is the dominant contributor in the full setting, thus excluding it produces a clear shift in the remaining clients. Food decreases from 0.3386 to 0.3209, Clothing decreases from 0.3298 to 0.3177, and Beauty increases from 0.3316 to 0.3614 by the last reported epoch. The growth of Beauty indicates that, in the absence of the primary knowledge carrier, the model leans on the domain with longer user sequences to stabilize cross-domain alignment. The three trajectories remain closer to each other than in the full setting, reflecting the loss of a strong semantic bridge once Kitchen is removed.

Food–Kitchen–Beauty (without Clothing). Food starts at 0.3329 and declines to 0.2920, Kitchen peaks early near 0.3861 and ends at 0.3471, while Beauty increases from 0.3274 to 0.3610. The dynamics show faster stabilization compared with other cases, which matches the observed early stopping in this setting. Since Clothing has a moderate number of exclusive users and shorter sequences than Beauty, its removal reduces diversity without severely harming cross-domain transfer. Beauty and Kitchen jointly anchor the shared representation, with Beauty becoming slightly dominant at convergence.

Food–Kitchen–Clothing (without Beauty). Kitchen grows from 0.3409 to 0.3763 and becomes the clear leader by epoch 25. Food trends upward modestly early on but ends at 0.3148, whereas Clothing ends at 0.3089. Excluding Beauty removes a domain with long sequences and less transferable overlap, which consolidates Kitchen’s role as the principal source of shared semantics. The resulting allocation mirrors the dataset sizes and overlap statistics, where Kitchen’s larger training set and central position in user co-occurrence provide the strongest cross-domain signal.

Taken together, the leave-one-out experiments produce consistent and interpretable patterns. When the domain that acts as a

semantic bridge is removed, the remaining clients reallocate influence toward the domains with longer sequences or larger training sets. When a peripheral domain is removed, the trajectories change little and the dominant contributor remains stable. These results

validate that the TracIn-based contribution evaluation is sensitive to the semantic structure of the federation and provides an explainable account of how each domain's shared knowledge shapes the global model.